

How many dimensions do I need?

Create a script to compute the number of feature dimensions **N** needed to represent at least 99.9% of the variance in the feature set of the humanactivity dataset using the 'pca' function.

The steps are:

- Compute **eigvals** using the 'pca' function
- Define vector **cumulative_percent_variance_permode**, which is a vector the same size as **eigvals** that contains 100 times (to convert fraction to percentage) the cumulative sum of the normalized eigenvalues
- Define **N** as the number of eigenvectors needed to capture at least 99.9% of the variation in our dataset **D**.

Script



Reset



MATLAB Documentation (<https://www.mathworks.com/help/>)

```
1 load humanact.mat
2 D = feat; % [24875 x 60] matrix containing 60 feature measurements from 24875 samples
3
4 % compute eigvals
5 [~,~eigvals] = pca(D);
6
7 % compute the cumulative_percent_variance_permode vector
8 cumulative_percent_variance_permode = 100*cumsum(eigvals) / sum(eigvals);
9
10 % Define N as the number of eigenvectors needed to capture at least 99.9% of the variation in D.
11 N = find(cumulative_percent_variance_permode >= 99.9, 1);
12
```

Run Script

Assessment: All Tests Passed

Submit

✔ Is eigvals correct
✔ Is cumulative_percent_variance_permode correct?
✔ Is N correct?

Output

Code ran without output.